Explorer Lincoln Ellsworth (left) and his copilot with their Northrop Gamma in the Antarctic in 1936

Great Manufacturers

Northrop

Consistently pushing the boundaries of aircraft design, John "Jack" Northrop maintained a visionary approach at the head of a series of companies. The emergence of the B-2 Spirit "Stealth Bomber," the most iconic aircraft to serve the US Air Force, was a result of his ambition to produce tailless flying wing aircraft.

BORN IN 1895 in New Jersey, USA, Jack Northrop was a capable and imaginative aircraft designer who worked for Douglas and Lockheed before cofounding his first company in 1927. The Avion Corporation was an outlet through which Northrop could pursue his radical design concepts. Throughout his life, Northrop's vision was to produce a flying wing aircraft—a tailless fixed-wing aircraft. He began work on this concept in 1929, but the program was put on hold during the Great Depression of the early 1930s. Around the same time, the Avion Corporation became part of the United Aircraft & Transport Corporation (UATC) and the first Northrop Corporation was established by Jack Northrop and Donald Douglas at El Segundo, California. Its earliest product was the Northrop Alpha. An all-metal construction, it was designed as a combined mail and passenger carrier and featured two significant innovations: a stressed-skin wing and wing fillets. Only 17 examples were built, but the influential Northrop Gamma, Beta, and Delta aircraft all emerged from this design. A Gamma called *Polar Star* was piloted by Herbert Hollick-Kenyon for the first transantarctic flight in 1935. The Gamma was developed in 1935 into the A-17 attack aircraft, which was the first Northrop aircraft to feature a retractable undercarriage. It entered service with the United States Army Air Corps (USAAC) in 1936.

John K. Northrop
(1895–1981)

In September 1937, the Northrop Corporation ceased trading. Two years later, Northrop and Moye Stephens formed Northrop Aircraft Incorporated at Hawthorne, California. Flying wing concepts were explored but, with war in Europe looming, they were dropped and efforts turned to building aircraft under licence, which supplied a useful source of income for the company.

In late 1940, the A-17-derived N-3PB Nomad emerged. Fitted with floats in place of wheels, it sold only to Norway, where it served as an anti-submarine aircraft and a convoy escort. Northrop's most successful production aircraft to date was the P-61 Black Widow. With a twin-boomed layout, the Black Widow was designed as a night interceptor fighter and proved extremely effective in service. Its success brought financial security and allowed Northrop to return to his vision.

In 1941, a concept flying wing design, the N-1M, was successfully tested. The follow-on N-9M was developed specifically as an interim step to the large bombers on Northrop's design boards. Northrop envisioned a series of long-range bombers. He believed they would be more efficient than traditional bombers, with a better all-round performance. Four N-9Ms were constructed and tested. The results of these tests fed into the XB-35 program, which had the USAAF's support. Early XB-35 test flights identified problems with the huge flying wing's engines. The US military ordered 13 YB-35s but the rapid development of jet-age technology rendered the propeller-driven aircraft obsolete. In order to resolve this, three YB-35s were converted into jet-powered YB-49s. Despite a successful start to the YB-49 program, the loss of two aircraft in accidents led to its cancellation and the flying wing concept was dropped.

P-61 Black Widow
One of the first aircraft specifically developed as a radar-equipped night fighter, the P-61 was introduced in 1944 and credited with the last kill of World War II.

Northrop A-17 attacks
This 1936 magazine *Popular Aviation* features the Northrop A-17 on its front cover. This craft was used by the USAF until 1944.

> "Now I know why God has **kept me alive** for the last twenty-five years."

JACK NORTHROP ON SEEING PLANS FOR THE B-2 SPIRIT, 1980